Hooke’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect spring displacement with restoring force.

SUCCESS CRITERIA - I can...

1. Measure displacement from the natural length.
2. Use spring constant in N/m.
3. Point the spring force back toward equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “spring constant” and “displacement” to explain your answer. Sentence starters are optional.

spring constant: how much an ideal spring’s force changes per stretch

displacement: change from initial position to final position, with direction

equilibrium: zero net force and zero acceleration

01 Warm-up: bring back an earlier idea

A point-mass pendulum drops **0.6 m** vertically from rest.
1. Use g=**10** and neglect drag.
2. **Find bottom speed**.

Word help: pendulum = an object that swings about a fixed support

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_{spring} = -kx$; $|F_{spring}| = k|x|$.
x is displacement from natural length; k is spring constant. The force restores the spring, within its linear range.

Worked reasoning: A spring with k=**100 N/m** stretches **0.03 m**, giving force magnitude kx=**3 N**. Its force on the attached object points toward equilibrium.

Your turn: Sketch equilibrium and stretched positions, then **draw the restoring-force arrow**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

A spring is stretched right.
1. **Draw the force it exerts** on the attached block.
2. **What changes** if compressed left?

Sentence starter: In my work, I used … to show … . I labeled it … .



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: 1 2 3

04 Apply the physics

A stationary **1 kg** load stretches a vertical spring **0.050 m**.
1. Use g = **9.8 N/kg**.
2. **Find the spring constant**, then **its force** magnitude at a **0.12 m** stretch.
3. Sketch and **label the load’s forces** for the stationary case.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Add and label force arrows.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says k doubles when the stretch doubles. **Explain which quantity actually doubles** for the same ideal spring.

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A force parallel to motion does **24 J** over **4 m**. **Find its size**. What work does the same force do with zero displacement?
Label force and displacement, including their relative direction.

Word help: parallel = along the same direction as a line or surface

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

07 Exit check

With right positive, k=**150 N/m** and x=**-0.08 m**, find signed spring force. **Mark the displacement** from equilibrium and **draw the restoring-force direction**.

Sentence starter: I found … [unit]. This value means … in this situation.



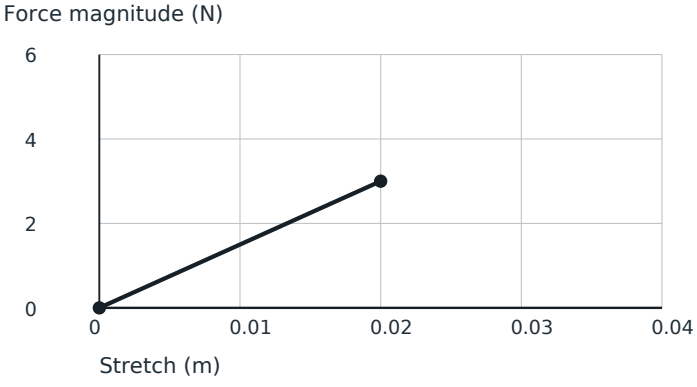
Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A spring graph passes through (x=**0.02 m**, F=**3 N**) and the origin. **Find k**, then **predict restoring force** at displacement x=**-0.04 m** using F=-kx. **Mark the displacement** from equilibrium and **draw the restoring-force direction**.

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: 1 2 3